

Future hyperloop testing infrastructure

Summary

The Hyperloop Development Program (HDP foundation), among other goals, aims to advance research and development in hyperloop technology via establishing testing infrastructures and pilot test tracks.

1. Current status of testing infrastructure in Europea

After a decade of international R&D, various hyperloop concepts and subsystems have been developed and tested. Currently, the goal is to maximize the value of current and planned R&D facilities (for roughly the next two years)

The objective of this phase is to achieve a harmonized preliminary system integration specification by utilizing existing and proposed European hyperloop R&D facilities. This stage focuses on demonstrating and de-risking the core operational aspects of hyperloop and aligning technology development with future standards. This phase is i.a. performed through the Hyper4Rail project (where HDP is the coordinator), in which final prototypes of key missing subsystems will be delivered, the overall concept harmonized, and the roadmap to industrialization created.

The main testing facilities in Europe currently are (non-exhaustive list, refer to WP5.1 Inventory of Already existing testing facilities.):

1. The existing European Hyperloop Center (EHC) in the Netherlands,
2. The DemoTube facility in Switzerland (under construction)
3. The realized TU Munich facility in Germany
4. The proposed Zeleros facility in Spain
5. Maglev test track by Nevomo
6. GoTube by Institute of Hyperloop Technology

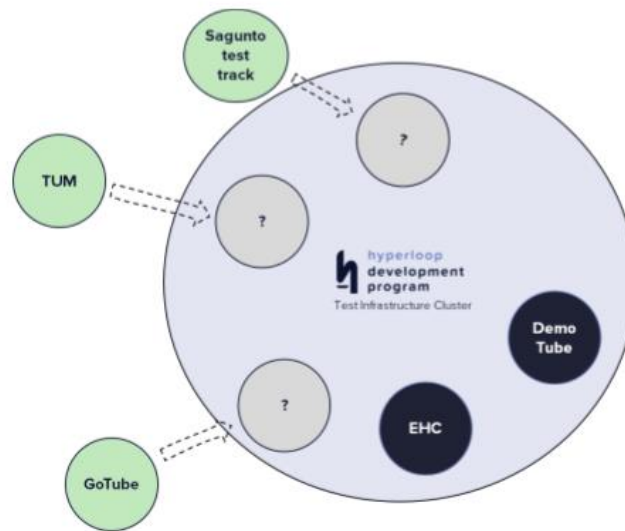
The HDP testing infrastructure cluster

In order to maximize the knowledge gained from these infrastructures, HDP has, together with the European hyperloop center (EHC) and EuroTube, started the “HDP testing infrastructure cluster”.

The collaboration between the HDP, EHC, and EuroTube offers significant benefits due to these key factors:

1. **Complementarity of test facilities:** While the European Hyperloop Center allows the testing of the vehicle design of hyperloop developers and the first hyperloop lane-switch, as well as the suitability of steel tubes, DemoTube enables the testing of the first hyperloop valve and the concept of sealed, post-tensioned concrete tubes.
2. **Resource Pooling and Efficiency:** HDP, EHC, and EuroTube are working closely together by pooling resources, increasing the effectiveness of both organizations. This collaboration constitutes the first steps towards creating a "European Space Agency" for hyperloop, with decentralized hyperloop testing sites throughout Europe coordinated by the HDP.

The collaboration aims to achieve the following goals:



Accelerated Innovation: The joint efforts are expected to significantly speed up the development and implementation of sustainable high-speed transport solutions in Europe and beyond, leading to advancements in the field of transportation technology.

Strengthen Europe's position: Through our combined efforts, we aim to position Europe at the forefront of hyperloop technology. This partnership will support Europe's commitment to innovative and sustainable transportation solutions.

Although the joining forces of EHC, HDP and EuroTube is a significant first step, the goal is to eventually incorporate as many test facility providers as possible.

2. Next steps needed (roll out plan)

2.a Full system prototype (Proof of Concept)

After the exhaustion of the test facilities that already exist or are under construction, a next step in scale will be needed, hereafter called Proof of Concept (POC). With the evidence of the parties' willingness to develop hyperloop independently, an introduction of a joint testing track to unify technology and foster standardization seems evident.

Currently, multiple entities across Europe are independently evaluating options to build their own test tracks:

- **EuroTube with AlphaTube:** EuroTube is developing the AlphaTube project, which aims to create a 3km test track in the Swiss Valais. This initiative focuses on providing a platform for comprehensive hyperloop technology testing and innovation at full speed.

- **TUM Seeking a New Site:** The Technical University of Munich (TUM) is actively seeking a new site to establish its own more-than-one-kilometer hyperloop test track. Their focus is on advancing academic research and development in the field of hyperloop technology.
- **Zeleros in Spain:** Zeleros, a leading hyperloop company based in Spain, is working on establishing a test track to validate their unique approach to their hyperloop vehicle, which emphasizes cost efficiency and integration with existing transportation infrastructure.

Potential Developments in Lathen, Germany: There are ongoing discussions about developing a hyperloop test track in Lathen, Germany. This site, previously used for magnetic levitation (maglev) train testing, offers existing infrastructure that could be adapted for hyperloop testing.

The main objective in this step is the physical construction of a 3-5 km 1:1 scale demonstrator at TRL7 (length requirement is given through full speed testing needs). By the end of this facility, the aim is to have collected sufficient empirical data from testing operations to inform publication of a 'draft for agreement' of the EN Standard. Indicatively, once the demonstrator is physically completed, this stage of the programme could be completed in 24 months.

2.b Potential future step: Minimum Viable System (4y+ timeline)

The last needed step before a network roll out could be the extension of the 3-5 km facility to form a 30-50 km twin tube Minimum Viable System (MVS) facility in which real-world, real-life hyperloop operations can be developed, demonstrated and certified for public service. Such an MVS is envisioned to have the following key characteristics:

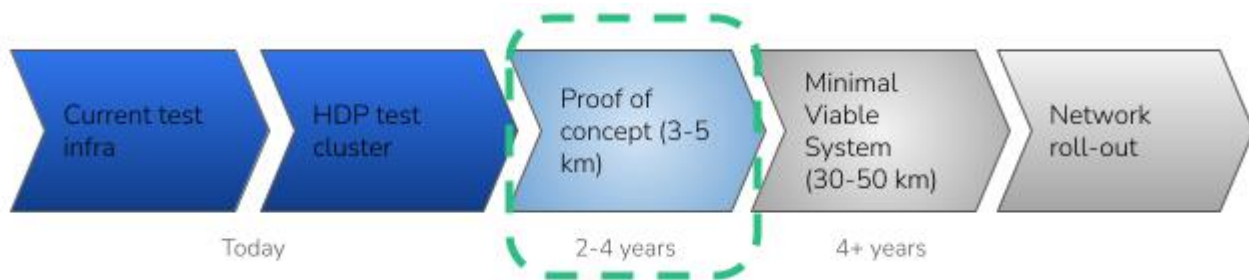
- Equipped from the outset for service-like passenger operations.
- Walk-on / walk-off stations at both ends of the route.
- Twin-tube infrastructure permitting bi-directional operations between both stations.
- A range of curves built in, to demonstrate g-compensation at operating speed.
- Switches at both ends of the route to permit vehicles to travel in both directions.
- High speed switch at the center point of the route, enabling demonstration of both single vehicle high speed switching, but also multi-vehicle 'splitting and joining' (if it is decided to adopt this principle).
- Capable of demonstrating at least 500 km/h (and ultimately up to 1000 km/h) maximum speed in human-in-vehicle operations.

The MVS could be realized either in a remote area where it can function as a certification facility for the long-term, or it could make sense to build MVS on a route where demand for high intensity transport exists for future revenue services. A major city center to a remote airport would be such an example.

By the end of this step, the aim is to have collected sufficient empirical data from POC and MVS operations to publish the definitive EN standards, by demonstrating reliable, robust, resilient high-speed hyperloop operation in service-like conditions over a 30-50 km route.

3. Realization of the full scale prototype (Proof of Concept)

The main work of this report is to outline how HDP could lead the way to a collaborative next generation test facility, here called Proof of Concept (PoC) as outlined in 2.a.



System requirements and deadlines

- **Budget Requirement:** Exceeding 200 million EUR (expected)
- **Track Length:** >3 kilometers, exact technical specs to be determined
- **Construction:** Only one test track will most likely be built
- **2-year timeline:** Startups must have a real project commitment in the coming years (2026-2028) to maintain investor support; failure to do so will result in loss of investment and potential closure of private hyperloop companies as shown with Virgin Hyperloop One.

Motivation: How to create a Win-Win Environment

To achieve this ambitious goal, it is crucial to converge efforts and resources towards developing a joint test track. The goal is not to convince HDP members (and ideally third parties) to change their current activities directly but to create an offer where they can continue their work with additional benefits. This approach will align their goals with ours, encouraging them to operate under the Hyperloop Development Project (HDP) umbrella, which is offering them a benefit.

How to do it - The “in-kind” approach

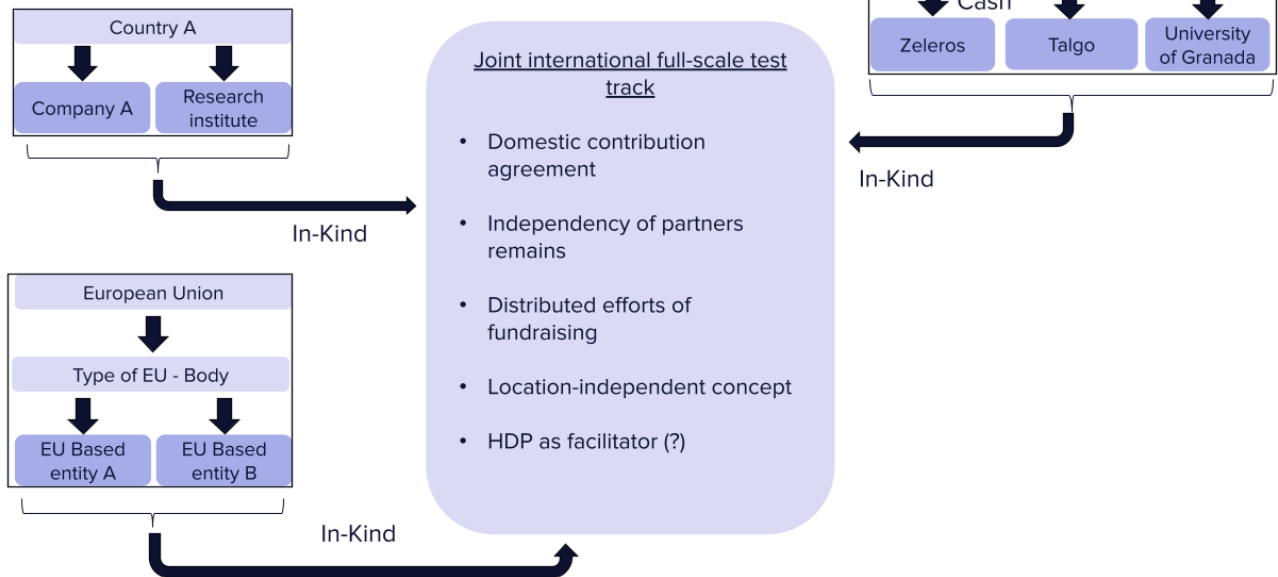
We have examined various centers financed by multiple countries (cf: Annexe 1), noting that the biggest benefit is knowledge sharing. A notable example is CERN, where researchers typically stay for a maximum of four years before returning to their home countries to share the acquired knowledge.

From looking at various such examples, a few key characteristics for such a PoC facility can be suggested:

Centralized Coordination: Centrally coordinated, ensuring that all participating countries contribute resources and expertise effectively. This central hub manages the overall direction, funding, and progress of the project.



POC ONE IDEA OF MANY : "IN-KIND APPROACH"



Industrial Focus: Focus on industrial applications rather than purely academic research ensures that its outputs are geared towards practical, commercially viable solutions.

Shared Funding: The Members deliver **little monetary contribution to the project**: instead, large parts of contributions will be delivered to the PoC entity in the form of completed components, systems or buildings through in-kind contributions (see example of ITER : in kind, cf Annexe 2)

Purchase Agreement Concept: The PoC structure could include a purchase agreement concept (similar to ITER, see cf Annexe 3), where participating countries commit to providing specific technologies, components, or services. These contributions are defined in detailed agreements that specify what each country will deliver, ensuring mutual benefit and commitment. Here is how it works :

1. **Identification of Needs:** The central coordinating body identifies the technologies, components, or services required for the project.
2. **Country Contributions:** Each participating country commits to supplying certain elements based on its industrial strengths and capabilities. For example, one country might provide advanced materials, while another supplies cutting-edge levitation systems.
3. **National Funding:** Countries use their national funding to develop these contributions domestically, supporting their local industries and technological ecosystems. This not only helps the project but also stimulates economic growth and innovation within each country.
4. **Delivery and Integration:** The developed technologies and components are then delivered to the central project site, where they are integrated into the overall system.

5. **In-Kind Contributions:** By contributing in-kind rather than purely financial contributions, countries ensure that their national industries benefit directly from the investment, fostering a symbiotic relationship between the national economies and the international project.

3. Next steps to realize the PoC test facility

On the HDP Partners Day 2024 in Zurich we **Introduced the in-kin Collaboration Model**. We presented the collaborative framework, emphasizing the potential for unlocking national funding and pooling resources into a common project. Generally, the approach was received well and to our impression, understood well.

As a next step, the following initiatives should be taken:

1. **Creation of Bilateral/Trial Agreements:** Establish initial agreements between Switzerland, the Netherlands, and possibly Spain.
2. **Engage with National Governments:** Begin discussions to secure government support and funding.
3. **Target Ambassadors and Economic Departments:** Focus on economic impact to gain backing from key governmental stakeholders.

Annexe 1 : Research institutions

Research Center	Location	Funding Structure	Funding Countries	Funding Percentages	Benefits
CERN	Geneva, Switzerland (partially in France)	23 member countries	Germany, France, UK, Italy, Spain, etc.	Approx. 20% EU, 80% member states	Access to research, Collaboration, Technological advancements, Educational/training, Industrial contracts
ITER	Cadarache, France	EU, China, India, Japan, Russia, South Korea, USA	EU, China, India, Japan, Russia, South Korea, USA	Approx. 45% EU, 55% other countries	Fusion research, Technological innovation, Skilled workforce, Experimental data, Long-term energy solutions
ESS	Lund, Sweden	13 European countries	Sweden, Denmark, Germany, UK, France, Italy, etc.	Approx. 35% Sweden, 65% other countries	Neutron research, Collaboration, Materials research, Scientific reputation
European XFEL	Hamburg and Schenefeld, Germany	12 European countries	Germany, France, Italy, Spain, Switzerland, etc.	Approx. 55% Germany, 45% other countries	X-ray research, Collaboration, Technological advancements, Educational/training, Shared IP
LLF	Marknesse, Netherlands	German-Dutch Wind Tunnels (DNW)	Germany, Netherlands	Approx. 50% Germany, 50% Netherlands	Aerodynamic testing, Collaboration, Technological advancements, Educational/training
ETW	Cologne, Germany	Germany, France, UK, Netherlands	Germany, France, UK, Netherlands	Approx. 40% Germany, 20% France, 20% UK, 20% Netherlands	Aerodynamic testing, Collaboration, Technological advancements, Industrial applications
Max Planck Society	Various locations in Germany	Germany, other European countries, EU	Germany, EU, other European countries	Approx. 70% Germany, 30% other countries	Cutting-edge research, Collaboration, Technological benefits, Training/development
ESRF	Grenoble, France	22 countries	France, Germany, UK, Italy, Spain, etc.	Approx. 25% France, 20%	Synchrotron research, Collaboration, Technological advancements, Educational/training, Shared IP

				Germany, 15% UK, 40% other countries	
JET	Culham, UK	EU, EURATOM program, associated countries	EU, EURATOM, associated countries	Approx. 80% EU, 20% other countries	Fusion research, Experimental data, Technological innovation, Skilled workforce, Energy solutions
EMBL	Heidelberg, Germany; other sites	27 member states	Germany, France, UK, Italy, Spain, etc.	Approx. 30% Germany, 70% other countries	Molecular biology research, Collaboration, Use of facilities, Educational/training, Technological applications
EMEC	Orkney, Scotland	Scottish Government, UK Government, EU, industry partners	UK, EU, industry partners	Approx. 30% UK, 50% EU, 20% industry partners	Marine energy testing, Collaboration, Technological advancements, Industrial applications, Shared research
ESTEC	Noordwijk, Netherlands	ESA member states	22 European countries	Approx. 40% Germany, 20% France, 20% Italy, 20% other countries	Space technology, Collaboration, Technological innovation, Educational/training, Industrial contracts
GSI Helmholtz Centre	Darmstadt, Germany	Germany, international partners	Finland, France, India, Poland, etc.	Approx. 60% Germany, 40% other countries	Heavy ion research, Collaboration, Technological innovations, Educational/training, Shared research
HFR	Petten, Netherlands	EU, NRG	EU, Netherlands, international collaborations	Approx. 50% EU, 50% Netherlands	Nuclear research, Collaboration, Technological advancements, Shared research, Industrial applications
ILL	Grenoble, France	France, Germany, UK	France, Germany, UK, other European countries	Approx. 30% France, 30% Germany, 30%	Neutron research, Collaboration, Technological advancements, Educational/training, Shared IP

				UK, 10% other countries	
ECMWF	Reading, UK	34 member and cooperating states	European countries	Approx. 60% EU, 40% other countries	Weather prediction, Collaboration, Improved forecasting, Educational/training, Shared research outputs
EMBL-EBI	Hinxton, UK	EMBL member states	27 member states	Approx. 30% Germany, 70% other countries	Bioinformatics research, Collaboration, Technological advancements, Educational/training, Shared IP
European Space Astronomy Centre (ESAC)	Madrid, Spain	ESA member states	ESA member countries	Approx. 40% Germany, 20% France, 20% Italy, 20% other countries	Access to research and facilities, Collaboration, Technological advancements, Educational and training opportunities
European Geostationary Navigation Overlay Service (EGNOS)	Toulouse, France	EU and ESA	EU, ESA member countries	Approx. 50% EU, 50% ESA	Access to research and facilities, Collaboration, Technological advancements, Educational and training opportunities
European Space Operations Centre (ESOC)	Darmstadt, Germany	ESA member states	ESA member countries	Approx. 40% Germany, 20% France, 20% Italy, 20% other countries	Access to research and facilities, Collaboration, Technological advancements, Educational and training opportunities
European Laboratory for Particle Physics (ELBE)	Dresden, Germany	German government, international collaboration	Germany, international partners	Approx. 60% Germany, 40% other countries	Access to research and facilities, Collaboration, Technological advancements, Educational and training opportunities

CERN Axion Solar Telescope (CAST)	Geneva, Switzerland (part of CERN)	CERN member countries	CERN member countries	Approx. 20% EU, 80% member states	Access to research and facilities, Collaboration, Technological advancements, Educational and training opportunities
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Annexe 2 : ITER Annual report - Financial data [link](#)

CONTRIBUTIONS RECEIVED FROM MEMBERS IN CASH

Amounts in thousands of Euro

Members	2022	Cumulative
EURATOM	169,428	2,044,702
PEOPLE'S REPUBLIC OF CHINA	55,791	393,993
REPUBLIC OF INDIA	81,516	296,237
JAPAN	44,370	377,841
REPUBLIC OF KOREA	58,266	399,261
RUSSIAN FEDERATION	58,247	411,474
UNITED STATES OF AMERICA	143,969	442,980
TOTAL CONTRIBUTIONS	611,587	4,366,489

CONTRIBUTIONS RECEIVED FROM MEMBERS IN KIND

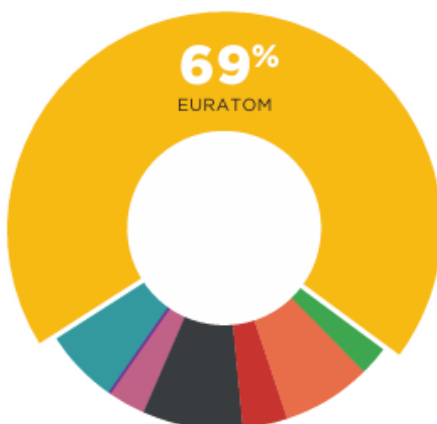
Amounts in ITER Unit of Account (IUA)

	Cumulative 31.12.2021	2022	Cumulative 31.12.2022
EURATOM	500,129	68,971	569,100
PEOPLE'S REPUBLIC OF CHINA	143,741	37,467	181,208
REPUBLIC OF INDIA	153,566	17,528	171,094
JAPAN	337,824	14,097	351,921
REPUBLIC OF KOREA	147,406	11,685	159,091
RUSSIAN FEDERATION	113,295	15,240	128,536
UNITED STATES OF AMERICA	88,725	10,139	98,864
TOTAL CONTRIBUTIONS	1,484,686*	175,128	1,659,814

* Due to a change in presentation, this 2021 cumulative total excludes accrued revenues. This explains the difference with the 2021 figure published in the 2021 ITER Organization Annual Report.

STAFFING AND FINANCIAL DATA - 45

STAFF BY MEMBER



	31/12 2018	31/12 2019	31/12 2020	31/12 2021	31/12 2022
CHINA	79	87	89	88	87
EURATOM	599	634	678	719	738
INDIA	36	28	28	22	26
JAPAN	27	32	35	37	38
KOREA	32	49	51	55	65
RUSSIA	37	47	59	68	77
USA	48	52	49	46	36
OTHER*					2
TOTAL	858 ⁽¹⁾	929 ⁽²⁾	989 ⁽³⁾	1,035 ⁽⁴⁾	1,069 ⁽⁵⁾

(1) Includes 5 Monaco Postdoctoral Fellows and staff funded for work on the Tokamak Cooling Water System (TCWS, 25), vacuum system (VAS, 2) and the safety control system for nuclear (SCS-N, 1)

(2) Includes 5 Monaco Postdoctoral Fellows and staff funded for work on the TCWS (29), VAS (3) and SCS-N (1)

(3) Includes 7 Monaco Postdoctoral Fellows and staff funded for work on the TCWS (27), VAS (2) and SCS-N (2)

(4) Includes 5 Monaco Postdoctoral Fellows and staff funded for work on the TCWS (25), VAS (2) and SCS-N (2)

(5) Includes 9 Postdoctoral Fellows (6 Monaco and 3 Korea) and staff funded for work on the TCWS (19), VAS (2) and SCS-N (2)

* "Other" refers to one Swiss and one Ukrainian staff member following ITER Council consultation and approval.

Annexe 3 : ITER Annual report - China and Japan purchasing agreement and credit allocation [link](#)



ITER CHINA (CN-DA)

www.iterchina.cn

PROCUREMENT ARRANGEMENTS*

Fourteen PAs signed since 2007 representing:

100% in number and
100% of the total value of CN-DA in-kind contributions.

Over 87 design or fabrication contracts related to ITER procurement have been signed with laboratories and industry.

Chinese procurement highlights in 2022

MAGNET SYSTEMS

TOROIDAL FIELD CONDUCTOR

All conductor unit lengths delivered

	% of ITER system procured by China	% Credit released out of allocation
TOROIDAL FIELD CONDUCTOR	7.5%	100%
POLOIDAL FIELD CONDUCTOR	65.2%	100%
MAGNET SUPPORTS	100%	45.93%
FEEDERS	100%	13.76%
CORRECTION COILS	100%	19.42%
CORRECTION COIL AND FEEDER CONDUCTORS	100%	100%

POLOIDAL FIELD CONDUCTOR

All conductor unit lengths delivered

MAGNET SUPPORTS

FEEDERS

Manufacture completed for all main busbar and correction coil high-temperature superconductor current leads

7 feeder segments and 1 batch of auxiliary hardware delivered this year to IO

Cumulative: 24 coil terminal boxes, 24 cryostat feedthroughs, 7 in-cryostat feeders (highly exceptional loads), 31 satellites, 3 batches of auxiliary hardware, and all bellows delivered

CORRECTION COILS

CORRECTION COIL AND FEEDER CONDUCTORS

All correction coil and feeder conductors delivered

POWER SYSTEMS

PULSED POWER ELECTRICAL NETWORK (PPEN)

All components of PPEN sub-package delivered

AC/DC CONVERTERS

All AC/DC converters for First Plasma delivered

REACTIVE POWER COMPENSATION

All reactive power compensation equipment installed on site; commissioning phase initiated

BLANKET

BLANKET FIRST WALL

Enhanced heat flux full-scale prototype assembled for process qualification

BLANKET SHIELD BLOCK

40% of manufacturing and testing has been completed

FUEL CYCLE

GAS INJECTION SYSTEM

FDR held successfully for the fusion power shutdown system (via video)

MRR held successfully for mass flow controlled valves (via video)

GLOW DISCHARGE CLEANING

DIAGNOSTICS

DIAGNOSTICS

FDR closure report approved for X-ray radial camera

MRR (1A closure report) approved for EQ#12 port integration

PDR closed for divertor Langmuir probe

Maturity assessment for all neutron flux monitor I&C (PDR design) completed

*Does not include Complementary Diagnostic Arrangements

Abbreviations : CDR Conceptual Design Review ; DA Domestic Agency ; FDR Final Design Review ; I&C Instrumentation & Control ; IO ITER Organization ; IC ITER Council ; FAT Factory Acceptance Tests ; MIP Manufacturing & Inspection Plan ; MRR Manufacturing Readiness Review ; PA Procurement Arrangement ; PDR Preliminary Design Review ; SAT Site Acceptance Tests



ITER JAPAN (JA-DA)

www.fusion.qst.go.jp/ITER/english/iter.html

PROCUREMENT ARRANGEMENTS*

Fifteen PAs signed since 2007 representing:

83% in number and

98% of the total value of JA-DA in-kind contributions.

Over 100 design or 800 design or fabrication contracts related to ITER procurement have been signed with industry since 2007.

Japanese procurement highlights in 2022

MAGNET SYSTEMS

TOROIDAL FIELD CONDUCTOR

All conductor unit lengths delivered

TOROIDAL FIELD MAGNET WINDINGS (9 OUT OF 19)

Toroidal field coil TF16 delivered to IO

TF15 is in transit, expected at IO in January 2023

Remaining two coils are under final machining

TOROIDAL FIELD MAGNET STRUCTURES

Delivery of magnet structures completed in 2022

CENTRAL SOLENOID CONDUCTOR

All conductor unit lengths delivered

HEATING & CURRENT DRIVE SYSTEMS

HIGH VOLTAGE BUSHING (MITICA)

Procurement of high voltage bushing completed

NEUTRAL BEAM HIGH VOLTAGE BUSHINGS

NEUTRAL BEAM BEAMSOURCE (HNB2 ACCELERATOR)

NEUTRAL BEAM POWER SUPPLY

PA signed in July for Heating Neutral Beam (HNB) Power Supply

ELECTRON CYCLOTRON RADIO FREQUENCY POWER SOURCES (8 GYROTRONS OUT OF 24)

First two gyrotrons (#1 and #2) delivered to IO with four superconducting magnets

All anode/body power supply sets delivered to IO and installation has started

All control cubicle sets delivered to IO and installation has started

FAT completed for gyrotron #6

ELECTRON CYCLOTRON EQUATORIAL LAUNCHER

Prototype of steering mirror assembly and waveguide successfully finished; final analysis underway to prepare for FDR

REMOTE HANDLING

BLANKET REMOTE HANDLING SYSTEM

Final design launched for main component

DIVERTOR

OUTER TARGET

Manufacturing of outer target prototypes in progress

Commissioning of Hot Helium Leak Test Facility underway

MRR for first six outer targets

Series manufacturing launched for parts of first six outer targets

TRITIUM PLANT

ATMOSPHERE DETRITIATION SYSTEM

Qualification activities in Japan completed

JA-DA/IO procurement activities proceeding

DIAGNOSTICS

DIAGNOSTICS

Microfission chamber: delivery of in-vessel components to IO

Microfission chamber: final design underway for other components

Edge Thomson scattering: FDR held for beam dump; final design of other components underway

Polaroid polarimeter: retroreflector in manufacturing; FDR held for equatorial port #10; final design for other components

Infrared thermography: final design in progress

Divertor impurity monitor and lower port #2 integration: preliminary design in progress

*Includes Complementary Diagnostic Arrangements

Abbreviations: CDR Conceptual Design Review; DA Domestic Agency; FDR Final Design Review; I&C Instrumentation & Control; IO ITER Organization; IC ITER Council; FAT Factory Acceptance Tests; MIP Manufacturing & Inspection Plan; MRR Manufacturing Readiness Review; PA Procurement Arrangement; PDR Preliminary Design Review; SAT Site Acceptance Tests.

50 - DOMESTIC AGENCY PROCUREMENT HIGHLIGHTS

Future hyperloop testing infrastructure - 2024 - Hyperloop Development Program